

Statistical Method Assignment IV

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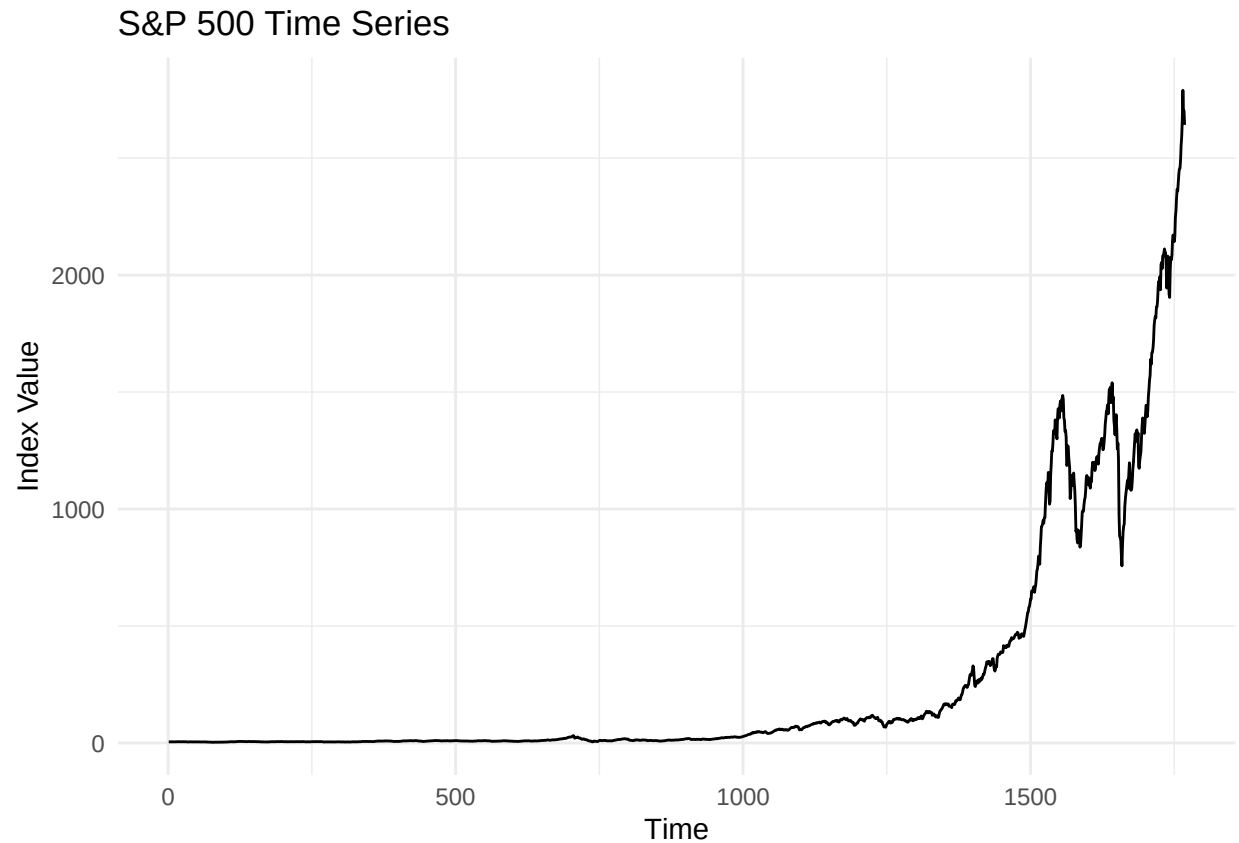
Problem 3

```
#Load Packages
library(lubridate)
library(forecast)
library(tseries)
library(ggplot2)

# Loading Data
AssignmentIV <- read.delim("~/R for Statistics and Data Science (STAT-5521G)/AssignmentIV.txt")

AssignmentIV$SP500 <- ts(AssignmentIV$SP500)

autoplot(AssignmentIV$SP500) +
  ggtitle("S&P 500 Time Series") +
  labs(x = "Time", y = "Index Value") +
  theme_minimal()
```

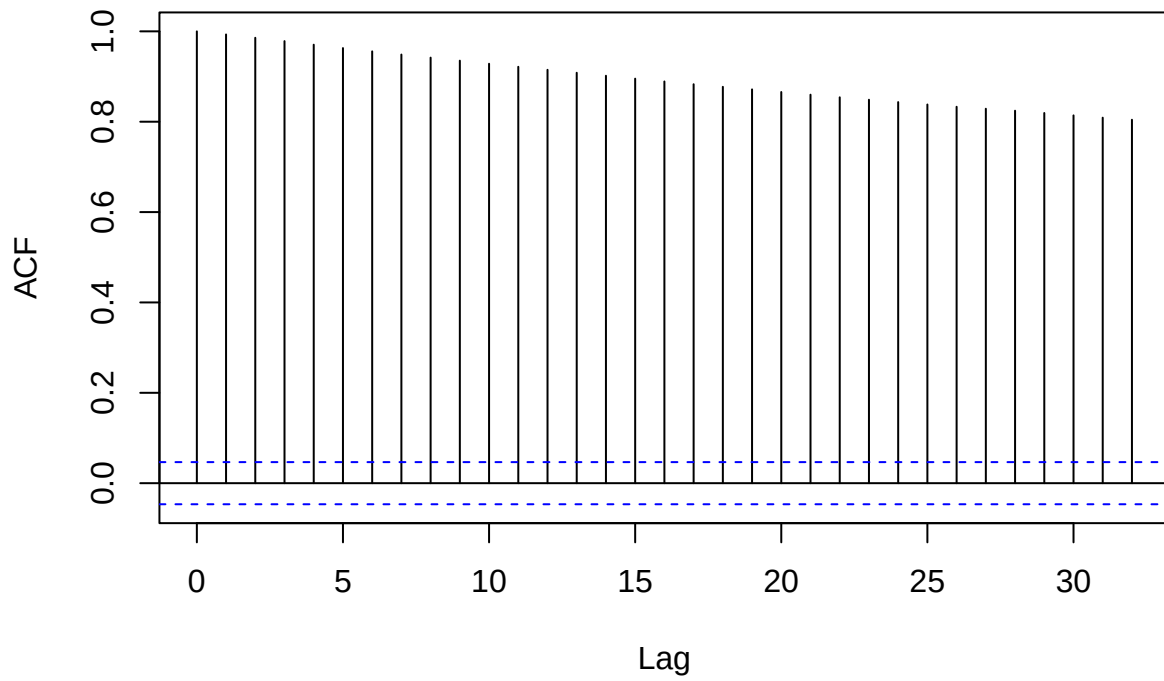


1. Does the data look stationary? Provide the R output. Do you need any transformation if you do provide an R output for it? (5pts).

Null Hypothesis: The time series is non-stationary. **Alternative Hypothesis:** The time series is stationary.

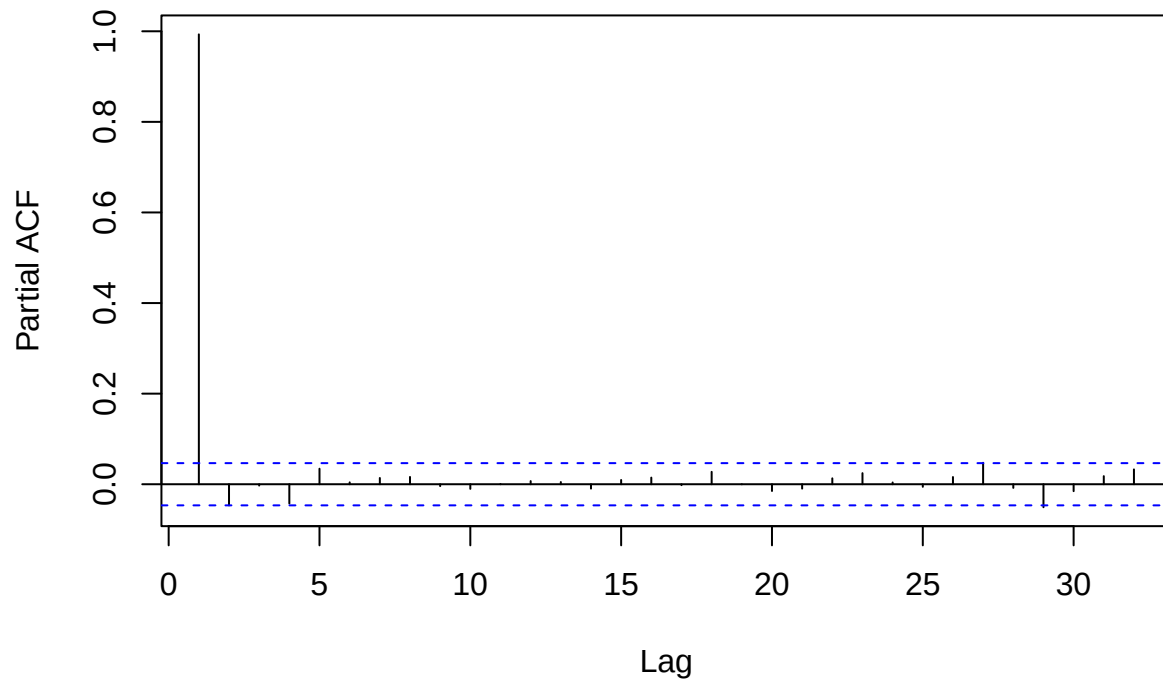
```
# Plot ACF and PACF  
acf(AssignmentIV$SP500, main = "ACF of SP500")
```

ACF of SP500



```
pacf(AssignmentIV$SP500, main = "PACF of SP500")
```

PACF of SP500



```
# Perform Augmented Dickey-Fuller Test  
adf.test(AssignmentIV$SP500)
```

```
##  
## Augmented Dickey-Fuller Test  
##  
## data: AssignmentIV$SP500  
## Dickey-Fuller = 1.6221, Lag order = 12, p-value = 0.99  
## alternative hypothesis: stationary
```

Decision and Conclusion: Since the p-value associated with the Dickey-Fuller test is greater than 0.05, we have all evidence to conclude that the data is non-stationary. Therefore, transformation and differencing of the data is inevitable.

2. Provide an R output for test for stationarity. (5pts)

```
#Transformation and Differencing  
diff_sp <- diff(AssignmentIV$SP500)  
  
# Test of Stationary  
adf.test(diff_sp)
```

```
##
```

```
## Augmented Dickey-Fuller Test
##
## data: diff_sp
## Dickey-Fuller = -9.7504, Lag order = 12, p-value = 0.01
## alternative hypothesis: stationary
```

Decision and Conclusion: Since the p-value associated with the Dickey-Fuller test is lesser than 0.05 level of significance, we have all evidence to conclude that the data is stationary. Therefore, we can go ahead to modeling our data.

3. What is the order of the ARIMA model? Provide the R output of how did you find it.(10pts)

```
model <- arima(diff_sp, order = c(1,1,3))
summary(model)
```

```
##
## Call:
## arima(x = diff_sp, order = c(1, 1, 3))
##
## Coefficients:
##          ar1      ma1      ma2      ma3
##      -0.6169  -0.1644  -0.7313  -0.0823
## s.e.    0.1411   0.1441   0.1019   0.0473
##
## sigma^2 estimated as 319.8:  log likelihood = -7600.39,  aic = 15210.79
##
## Training set error measures:
##              ME      RMSE      MAE MPE MAPE      MASE      ACF1
## Training set 0.4684965 17.87708 6.276823 NaN  Inf  0.7679236 -0.001092408
```

4. Report the theoretical model with the corresponding parameters. (5pts)

ARIMA(1,1,3)

This implies the model can be written in its theoretical form as:

$$(1 - \phi_1 B)(1 - B)y_t = (1 + \theta_1 B + \theta_2 B^2 + \theta_3 B^3)\varepsilon_t$$

Where:

- y_t is the S&P 500 index at time t
- B is the backshift operator
- ϕ_1 is the autoregressive (AR) coefficient
- $\theta_1, \theta_2, \theta_3$ are the moving average (MA) coefficients
- ε_t is the white noise error term

So, the final theoretical ARIMA model is:

$$(1 + 0.6169B)(1 - B)y_t = (1 - 0.1644B - 0.7313B^2 - 0.0823B^3)\varepsilon_t$$

5. Provide an estimated values for the parameters, using R and the fitted model output. (10pts)

```
model$coef
```

```
##          ar1          ma1          ma2          ma3  
## -0.61685871 -0.16437017 -0.73129375 -0.08226873
```

6. Provide a forecast plot for the model using R. (5 pts)

```
autoplot(forecast(model))+  
  labs(x = "Time", y = "Index Value")+  
  theme_minimal()
```

